Data Semantics - Assignment 3 - Generate with Transformers

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Overview

This assignment focuses on autoregressive generative models. You will work directly with the `openai-community/gpt2` model and manually implement a `generate` function. This function will generate tokens one at a time, autoregressively, either until a predefined maximum number of new tokens has been generated, or until a stop token is encountered.

You are provided with a Jupyter notebook[1] that contains the base code required for the exercises. The notebook includes useful hints, partial implementations, and expected outputs to guide your work.

You may choose to run the notebook in the cloud using Google Colab (suggested) or locally by installing Jupyter[2].

Questions and Exercises

Run the first cells of the provided notebook and answer the first questions. The continue with the exercises and the final question.

1. What does `output.logits` represent?
2. What does `output.logits[0, -1, :]` represent?
3. Implement the greedy generation strategy by completing the `greedy_generate` function. Greedy decoding should always choose the most probable next token at each step. Refer to the notebook for expected output.
4. Extend the generation function to support random sampling from the token distribution, instead of always picking the most probable token.
5. Implement `top_k` sampling, where the next token is randomly selected from the top-k most probable tokens.

[1] `https://drive.google.com/file/d/161RE65yuNG1PlKYyg9FIKCO8dJRi6Y1G/view?usp=sharing`

[2] `https://jupyter.org/install`

6. Introduce and implement *temperature scaling* in the sampling procedure.

   Temperature controls the randomness of the token distribution during generation. Given logits z_i, we define the **temperature-adjusted softmax** as:

$$\text{Temperature-adjusted Softmax:} \quad p_i = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}$$

   Extend your sampling implementation to support a temperature parameter, and modify the token selection accordingly. Experiment with different values of T (e.g., 0.7, 1.0, 1.5).

7. Does it make sense to apply temperature scaling in greedy decoding?